

(*) Let A be any set.

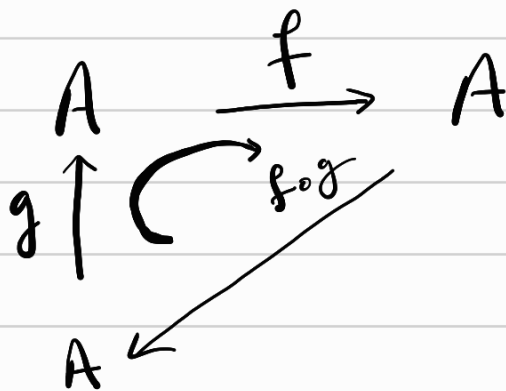
$$G = \{ f: A \rightarrow A \mid f \text{ is a bijection } \}$$

Prove that $(G, \circ)$ is a group.

pf: ① closure: let f, g $\in G$

we need, fog $\in G$

as f, g are bijective fun,



fog is also bijective.

We first show that f o g is 1-1:

$$\underline{f \circ g}(x) = \underline{f \circ g}(y) \quad \text{for } x, y \in A$$

$$\Rightarrow \underline{f}(\underline{g}(x)) = \underline{f}(\underline{g}(y)) \quad , \quad \underline{f} \text{ is bijective} \Rightarrow \underline{f} \text{ is 1-1}$$

$$\Rightarrow \underline{g}(x) = \underline{g}(y)$$

$$\Rightarrow \underline{x} = \underline{y} \Rightarrow \underline{f \circ g} \text{ is 1-1}$$

(*) fog is onto:

$f \circ g : A \rightarrow A$ f is bijective $\Rightarrow f$ is onto.

Let $y \in A$,
From (I)

$$f(x) = y$$

$$\Rightarrow f(g(x)) = y \text{ [from (II)]}$$

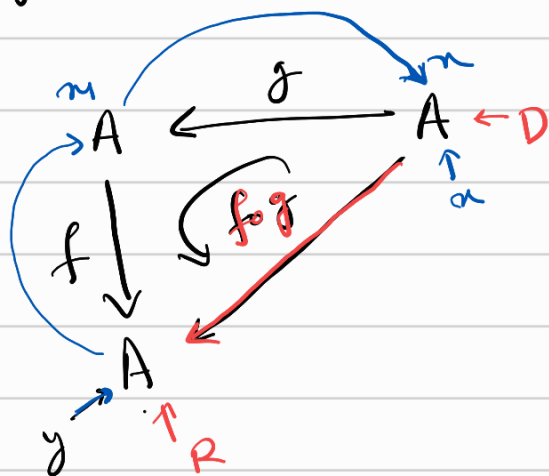
$$\Rightarrow f \circ g(x) = y.$$

$$\exists x_1 \in A \text{ s.t. } f(x_1) = y. \text{ --- (I)}$$

$$f : A \rightarrow A$$

$$g : A \rightarrow A$$

$$g(x) = x_1 \text{ --- (II)}$$



$$(f \circ g)^{-1} = g^{-1} \circ f^{-1}$$

$$\Rightarrow f \circ g \in G$$

$$(II) \quad f, g, h \in G, \quad f \circ (g \circ h) = (f \circ g) \circ h. \text{ (why?)}$$

$$\forall x \in A, \quad f \circ (g \circ h)(x) = f((g \circ h)(x))$$

$$= f(g(h(x)))$$

$$(f \circ g) \circ h(x) = (f \circ g)(h(x))$$

$$= f(g(h(x)))$$

$$\Rightarrow f \circ (g \circ h) = (f \circ g) \circ h$$

$$(III) \quad Id_A : A \rightarrow A$$

$$x \mapsto x$$

$$Id_A(x) = x, \quad Id_A \in G.$$

we want to show $f \circ \text{Id}_A = f$

$$\forall x \in A, \underline{f \circ \text{Id}_A}(x) = f(\underbrace{\text{Id}_A(x)}_x) = \underline{f(x)}$$

$$\Rightarrow \underline{f \circ \text{Id}_A = f}$$

$$(\text{Id}_A \circ f)(x) = \underline{\text{Id}_A(f(x))} = f(x)$$

$$\Rightarrow \boxed{\text{Id}_A \circ f = f = f \circ \text{Id}_A}$$

(IV) f bijective map, $\exists f^{-1} : A \rightarrow A$.

$$f \circ f^{-1}(x) = f(\underline{f^{-1}(x)}) = x = \text{Id}_A(x)$$

$$\Rightarrow f \circ f^{-1} = \text{Id}_A$$

$$f^{-1} \circ f(x) = f^{-1}(f(x)) = x = \text{Id}_A(x)$$

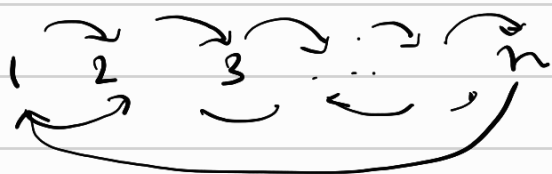
$$\Rightarrow \boxed{f^{-1} \circ f = \text{Id}_A}$$

$\left(G_2 = \left\{ f: \underset{\substack{\downarrow \\ \text{any set}}}{A} \rightarrow A \mid f \text{ is bijection} \right\}, \circ \right)$ is a group.

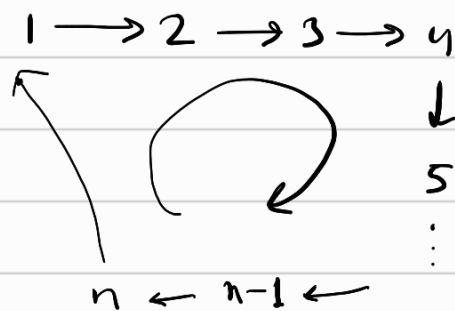
• particularly, $A = \{1, 2, 3, \dots, n\} = \{a_1, a_2, \dots, a_n\}$

$\sim G_2 = \left\{ f: A \rightarrow A \mid f \text{ is bijection} \right\} = \underline{S_n}$ (Symmetric group / permutation group on n element set)

• $\text{order}(S_n) = |S_n| = o(S_n) = n!$



$$\sigma \in S_n \quad \begin{aligned} \sigma(1) &= 2 \\ \sigma(2) &= 3 \\ &\vdots \\ \sigma(n) &= 1 \end{aligned}$$



$$\sigma = \begin{pmatrix} 1 & 2 & 3 & \dots & n-1 & n \\ 2 & 3 & 4 & \dots & n & 1 \end{pmatrix}$$

$$\rho = \begin{pmatrix} 1 & 2 & 3 & 4 & \dots & n-1 & n \\ 2 & 1 & 4 & 3 & \dots & n & n-1 \end{pmatrix} \rightarrow n \text{ even}$$

$$S_3, \rho = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \end{pmatrix}$$

$$\rho_1 = \begin{pmatrix} 1 & 2 & 3 \\ 1 & 2 & 3 \end{pmatrix} = \rho_1^{-1}$$

$$\rho_2 = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 1 \end{pmatrix}$$

$$\rho_2^{-1} = \begin{pmatrix} 1 & 2 & 3 \\ 3 & 1 & 2 \end{pmatrix}$$

$$\begin{aligned} \rho_2(1) &= 2 \Rightarrow \rho_2^{-1}(2) = 1 \\ \rho_2(2) &= 3 \Rightarrow \rho_2^{-1}(3) = 2 \end{aligned}$$

$$\downarrow P_2(3) = 1 \Rightarrow P_2^{-1}(1) = 3$$

$$\begin{pmatrix} 1 & 2 & 3 \\ 3 & 1 & 2 \end{pmatrix}$$

$$P_2 = \begin{pmatrix} 1 & 2 & 3 \\ \uparrow & \uparrow & \uparrow \\ 2 & 3 & 1 \end{pmatrix}$$

$$P_2^{-1} = \begin{pmatrix} 2 & 3 & 1 \\ 1 & 2 & 3 \end{pmatrix}$$

$$= \begin{pmatrix} 1 & 2 & 3 \\ 3 & 1 & 2 \end{pmatrix}$$

$$P_3 = \begin{pmatrix} 1 & 2 & 3 \\ 3 & 2 & 1 \end{pmatrix} \begin{matrix} \uparrow \\ \downarrow \end{matrix}$$

$$P_3^{-1} = \begin{pmatrix} 3 & 2 & 1 \\ 1 & 2 & 3 \end{pmatrix}$$

$$= \begin{pmatrix} 1 & 2 & 3 \\ 3 & 2 & 1 \end{pmatrix}$$

$$P_4 = \begin{pmatrix} 1 & 2 & 3 \\ 1 & 3 & 2 \end{pmatrix} \begin{matrix} \uparrow 2 \\ \downarrow 3 \end{matrix} P_4^2 =$$

$$P_4^{-1} = \begin{pmatrix} 1 & 3 & 2 \\ 1 & 2 & 3 \end{pmatrix}$$

$$= \begin{pmatrix} 1 & 2 & 3 \\ 1 & 3 & 2 \end{pmatrix}$$

$$P_6 = \begin{pmatrix} 1 & 2 & 3 \\ 3 & 1 & 2 \end{pmatrix} \begin{matrix} 1 \rightarrow 3 \\ 3 \rightarrow 2 \\ 2 \rightarrow 1 \end{matrix}$$

$$P_6^{-1} = \begin{pmatrix} 3 & 1 & 2 \\ 1 & 2 & 3 \end{pmatrix} \rightarrow P_6^3 = \text{Id}$$

$$= \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 1 \end{pmatrix}$$

Th^m: Let G be a group and $g \in G$. Then $o(g) = o(g^{-1})$.

pf: Let $o(g) = n$, $o(g^{-1}) = m$, claim: $n = m$.

$$(g^{-1})^n = (g^n)^{-1} \quad [\text{From previous claim}]$$

$$\begin{aligned} (g^{-1})^m &= e = (e)^{-1} = e & (g^{-1})^n &= (g^n)^{-1} = e^{-1} = e \\ \Rightarrow (g^m)^{-1} &= e \Rightarrow g^m = e & \Rightarrow m &| n. \\ \Rightarrow g^n &= e & \Rightarrow n &| m. \\ \Rightarrow m &\geq n. & \Rightarrow n &\geq m. \\ \Rightarrow m &= n. \end{aligned}$$

Let $x \in G$, if $o(x) = n$, then $\forall k \in \mathbb{N}$ s.t. $x^k = e \Leftrightarrow n | k$

pf: $n | k \Rightarrow k = nq$, for some $q \in \mathbb{N}$.

$$x^k = x^{nq} = (x^n)^q = (e)^q = \underline{e}.$$

$$\Rightarrow x^k = e.$$

$$[n|k] \Rightarrow x^k = e$$

conversely, $(x^k = e)$

$d(x) = n \Rightarrow$ n is the smallest natural no such that

$$x^n = e, \Rightarrow \underline{k \geq n} \Rightarrow n|k \Leftrightarrow$$

$$\Rightarrow \boxed{k = nq + r} \quad \text{--- (1)}$$

Euclidean's Division Algorithm

$$\rightarrow \boxed{0 \leq r < n}$$

$$x^k = e$$

$$\Rightarrow x^{nq+r} = e \Rightarrow x^{nq} \cdot x^r = e$$

$$\Rightarrow (x^n)^q \cdot x^r = e$$

$$\Rightarrow e^q \cdot x^r = e$$

$$\Rightarrow \underline{x^r = e} \Rightarrow \underline{r=0}$$

$$\boxed{x^1 \cdot x^{-1} = x^0 = e}$$

From eqⁿ (1),

$$k = nq + 0 \Rightarrow k = nq \Rightarrow \boxed{n|k}$$

$$(\mathbb{R}, +, \cdot)$$

$$x = \frac{-a_1 \pm \sqrt{a_1^2 - 4a_2a_0}}{2a_2}$$

$a_2x^2 + a_1x + a_0 = 0$

$$(F, +, \cdot) \quad (E, +, \cdot)$$

$$a_3x^3 + a_2x^2 + a_1x + a_0 = 0$$

$$x = \dots$$

$+$, $\times$

$$a_n x^n + \dots + a_0 = 0$$

$$\rightarrow a_n x^n + \dots + a_0 = 0$$

$$n \geq 5$$

Group theory, Ring theory, Field theory

Order of a permutation:

Let $\sigma \in S_n$ be any permutation. Then order of σ is k iff $\sigma^k = \text{Id}$ where k is smallest natural no. ≥ 1 and Id identity permutation, $\text{Id}(x) = x \quad \forall x \in \{1, 2, \dots, n\}$

$$\text{Id} = \begin{pmatrix} 1 & 2 & 3 & 4 & \dots & n \\ 1 & 2 & 3 & 4 & \dots & n \end{pmatrix}$$

$$\sigma = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & \dots & n \\ 2 & 1 & 4 & 3 & 5 & 6 & \dots & n \end{pmatrix}$$

$$\Rightarrow \sigma^2 = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & \dots & n \\ 1 & 2 & 3 & 4 & 5 & 6 & \dots & n \end{pmatrix} = \text{Id}$$

$$\begin{aligned} \sigma^2(5) &= \sigma(\sigma(5)) = \sigma(5) \\ \sigma^2(2) &= \sigma(\sigma(2)) = \sigma(1) = 2 \\ \sigma^2(1) &= \sigma(\sigma(1)) = \sigma(2) = 1 \end{aligned}$$

Cycle:

$$\sigma_1 = \begin{pmatrix} \text{cycle } (1 \ 2) & \text{cycle } (3 \ 4) & 5 & 6 & \dots & n \\ 2 & 1 & 4 & 3 & 5 & 6 & \dots & n \end{pmatrix}$$

$$= \begin{matrix} 1 \rightarrow 2 \rightarrow 1 & 3 \rightarrow 4 \rightarrow 3 & 5 \rightarrow 5 & 6 \rightarrow 6 & \dots & n \rightarrow n \end{matrix}$$

length 2-cycle length 2-cycle

$\sigma_2 = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & \dots & n \\ 2 & 3 & 1 & 4 & 5 & 6 & \dots & n \end{pmatrix} = (1 \ 2 \ 3)$

$\sigma_2^3 = Id$

length 3 cycle

$4 \rightarrow 4, 5 \rightarrow 5, 6 \rightarrow 6, \dots, n \rightarrow n$

$\sigma_3 = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & \dots & n-1 & n \\ 2 & 3 & 4 & 5 & 6 & 7 & \dots & n & 1 \end{pmatrix}$

$\sigma_3 = (1 \ 2 \ 3 \dots n)$

n -length cycle

Observation: In S_n , any permutation has at most n -length cycle.

Observation: In S_n , only permutation with all cycles of length 1 is identity permutation.

Defⁿ: Identity permutation is a cycle of length 1.

$Id := (\overset{\vee}{1}) \text{ or } (\overset{\vee}{2}) \text{ or } (\overset{\vee}{3}) \text{ or } \dots \text{ or } (\overset{\vee}{n})$

defined as

Defⁿ: If in any permutation σ in S_n , $\exists$ only one cycle of a specific length (say $k, k \neq 1$) and rest are of length 1-cycle, then we say σ is a cycle of length k .

Let σ be a k length cycle:

$$k+2-3+1 \\ 1 \rightarrow k \quad k-1+1$$

$$\sigma = \begin{pmatrix} 1 & 2 & 3 & 4 & \dots & k-1 & k & k+1 & k+2 & \dots & n \\ 2 & 3 & 4 & 5 & \dots & k & k+1 & k+2 & 3 & \dots & n \end{pmatrix}$$

$$= (3 \ 4 \ 5 \ \dots \ k+2)$$

order of $\sigma = k$

Th^m: k length cycles have order k .

pf: $(1 \ 2 \ 3 \ \dots \ k)$ (construction proof)

Disjoint cycle: $\sigma = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & \dots & n \\ 2 & 1 & 4 & 3 & 5 & \dots & n \end{pmatrix}$

$(1 \ 2 \ 3)$
 $(1 \ 2)$ $(2 \ 1)$

$= (1 \ 2) (3 \ 4)$

$(1 \ 2 \ 3 \ 4 \ 5 \dots n)$
 $(1 \ 2 \ 4 \ 3 \ 5 \dots n)$

$\sigma_1 = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 1 \end{pmatrix} = (1 \ 2) (2 \ 3) (3 \ 1)$

$= (1 \ 2 \ 3)$

transposition.

$$(1 \ 2 \ 3 \ \dots \ k) = (1 \ 2)(2 \ 3)(3 \ 4) \dots (k-1 \ k) \underbrace{(k \ 1)}_{\text{K transposition}}$$

$$\sigma = (1 \ 2)(3 \ 4) = (3 \ 4)(1 \ 2) \quad [\text{Disjoint cycles commute}]$$

$$\sigma^2 = (1 \ 2)(3 \ 4)(1 \ 2)(3 \ 4)$$

$$= (1 \ 2)(1 \ 2)(3 \ 4)(3 \ 4)$$

$$= (1 \ 2)^2 (3 \ 4)^2$$

$$= Id \ Id = Id$$

****** Th^m: Every permutation $\sigma \in S_n$ can be written as product of 2 cycles (transposition).

pf: Case-I: σ is a k cycle.

$$\sigma = (1 \ 2 \ 3 \ \dots \ k) \quad (\text{wlog}) -$$

$$= (1 \ 2)(2 \ 3)(3 \ 4) \dots (k-1 \ k)$$

(aim: $\sigma = \{$

$$\sigma(\underline{i}) = \underline{i+1} \quad \forall \underline{i} \quad \Bigg| \quad \sigma(1) = 2$$

$\Leftrightarrow \sigma = \{$

$$\zeta(1) = (1\ 2)(2\ 3)(3\ 4) \dots (k-1\ k)$$

$$= (1\ 2)(2\ 3)(3\ 4) \dots (k-1\ k)(k)$$

$$= (1\ 2)(2\ 3)(3\ 4) \dots (k-1)$$

$$=$$

$$\begin{aligned} & f_1 \circ f_2 \circ f_3 \dots \circ f_n(i) \\ &= \underline{f_1 \circ f_2 \circ f_3} \circ \underline{f_n(i)} \end{aligned}$$

Ex:

$$\sigma = (1\ 2\ 3\ 4\ 5) \Rightarrow \sigma(1) = 2$$

$$\zeta = (1\ 2)(2\ 3)(3\ 4)(4\ 5)$$

$$(4\ 5) = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 1 & 2 & 3 & 5 & 4 \\ & & & 6 & 7 \dots \end{pmatrix}$$

$$\zeta(1) = (1\ 2)(4)$$

$$= 2$$

$$\zeta(2) = 3 = \sigma(2)$$

$$\zeta(3) = \sigma(3)$$

$$\boxed{\zeta = \sigma}$$

Cor-II:

Lemma: (Sub Theorem): Given any $\sigma \in S_n$, we can express σ as composition of disjoint cycles.

Ex:

$$\begin{aligned} \sigma &= \left(\begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix} \begin{pmatrix} 3 & 4 \\ 4 & 3 \end{pmatrix} \right) \in S_n \\ &= (1\ 2)(3\ 4) \end{aligned}$$

$$\sigma = \begin{pmatrix} \boxed{1 \ 2} & \boxed{3 \ 4} & \boxed{5 \ 6} \\ \boxed{2 \ 5} & \boxed{4 \ 3} & \boxed{6 \ 1} \end{pmatrix}$$

$$= \underbrace{(3 \ 4)}_{\uparrow} \circ \underbrace{(1 \ 2 \ 5 \ 6)}_{\uparrow} = (1 \ 2 \ 5 \ 6) \circ (3 \ 4)$$

Th^m: Disjoint cycles commute:

$$\sigma, \tau \in S_n, \sigma \cap \tau = \emptyset; \boxed{\sigma \circ \tau = \tau \circ \sigma}$$

Cor-II: $\sigma = \underbrace{\sigma_1 \circ \sigma_2 \circ \sigma_3 \cdots \sigma_\ell}_{\substack{\bigwedge_{i,j,k,l} \sigma_i \cap \sigma_j \cap \sigma_k \cap \sigma_l = \emptyset \\ \text{by above Lemma}}}, \sigma_i \cap \sigma_j = \emptyset \ \forall i \neq j$

For each σ_i apply Cor-I, $\sigma_i = \sigma_{i1} \circ \sigma_{i2} \cdots \sigma_{il_i}$

σ_{ij} is a transposition / 2-cycle.

Ex: $\sigma = (3 \ 4) (1 \ 2 \ 5 \ 6)$
 $= \boxed{(3 \ 4)} \boxed{(\underline{1 \ 2})(\underline{2 \ 5})(\underline{5 \ 6})}$

Observation: ① 2-cycle order is 2

② Disjoint cycles commute.

$$\sigma = (3 \ 4) (1 \ 2 \ 5 \ 6)$$

$$\sigma^2 = \sigma \circ \sigma = (3 \ 4) (\underline{1 \ 2 \ 5 \ 6}) (\underline{3 \ 4}) (\underline{1 \ 2 \ 5 \ 6})$$

$$= \underline{(3\ 4)} \underline{(3\ 4)} (1\ 2\ 5\ 4) (1\ 2\ 5\ 4)$$

$$= \underline{(1\ 2\ 5\ 4)} \underline{(1\ 2\ 5\ 4)} \quad \bigg| \quad (1\ 2\ 5\ 4)^4 (1) = 1$$

$$\sigma^3 = (1\ 2\ 5\ 4) (1\ 2\ 5\ 4) \overbrace{(3\ 4) (1\ 2\ 5\ 4)}$$

$$= (1\ 2\ 5\ 4)^3 (3\ 4)$$

$$\sigma^4 = (1\ 2\ 5\ 4)^3 \underline{(3\ 4) (3\ 4)} (1\ 2\ 5\ 4)$$

$$= (1\ 2\ 5\ 4)^4 = Id.$$

** Th^m: Order of $\sigma \in S_n$ is lcm of order of disjoint cycles.

Ex: $\sigma = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 2 & 3 & 1 & 5 & 4 \end{pmatrix}$

$$= \underbrace{(1\ 2\ 3)}_3 \underbrace{(4\ 5)}_2$$

$$o(\sigma) = \text{lcm}(3, 2) = 6.$$